

thuRsd day R Workshop

Lecture 1: Data – They are not just tables anymore

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What Data can R Deal With?

Most of us store data in a spreadsheet. We arrange various attributes or variables in columns and rows represent observations (also called samples or exemplars). While this is a common data format, it is not the only data format that we can analyze these days. For example, we can read a wide range of files such as:

- Images (tif, png, .jpeg,...)
- Videos (mp4, .gif,..)
- Spatial Datasets (shp, gpkg,...)
- Time series data (csv, .xls)
- Textual data (.txt)
- Special formats (.spss,...)

How does R Access your Data?

The process of R accessing your data for further analysis is called **reading the data**. R typically reads the data by:

- Reading the data on your hard-disk and copying it to the RAM
- By establishing a connection where in data on the hard-disk can be brought in as necessary.
- Any data that is in the RAM is erased when you shut down R unless it is saved to a hard-disk prior to shut down

Data Representations in R

Data representation indicate how R stores data internally. This is important for several reasons including - 1) How much memory is allocated to the data; 2) How data or its subsets can be accessed by the program.

Data types: denote what type of data can be stored in R. Basic data types include

- **Integer** – Whole numbers (0,1,2...)
- **Float** – Fractional numbers (0.1, 0.001, 1E10). They are usually stored as double precision for greater accuracy
- **Boolean** – TRUE or FALSE. Internally they may be stored as 1 or 0
- **Factors** – Used to denote categories (Low, Medium, High) – These are stored internally as numbers but can be used as factors
- **Strings** – Text information that are stored within single or double quotes
- **Complex** – Used to store complex numbers ($a + bi$) where a is the real part and b is the imaginary part (useful for engineering and mathematical calculations)

Data Structures: These are used to store sequences of data. Data structures tell R how data should be stored and retrieved from the computer memory into the program. R data structures can be divided into two categories

Homogeneous data structures: These are used to store data sequences, all of which have one data type. The 3 basic homogeneous data structures are:

- **Atomic Vectors or Vectors:** denoted by `c(1.2,2.3,4.2)` or `c('A', 'B', 'C')`.
- **Matrix:** A two-dimensional representation of homogeneous data. A matrix can also be thought of as a vector of vectors. The elements of a matrix `M` is denoted by two indices `(r, c)` where `r` is the row and `c` is the column

$A = \begin{pmatrix} 1.2 & 2.3 & 4.6 \\ 1.8 & 5.2 & 3.1 \\ 3.6 & 3.7 & 7.5 \end{pmatrix}$ is a 3 x 3 matrix. The element `A[2,3] = 3.1` and `A[3,2] = 3.7`

- **Array:** An array is a stack of matrices or a vector of matrices. An element of an array is denoted by `A[r,c,i]` where `r` is the row, `c` is the column and `i` is the index (stack position) of the matrix

Heterogeneous Data Structures: As their name suggests heterogeneous data structures can store different types of data. There are two basic data structures

- **List:** denoted by `list("a",2,3+2i,FALSE)` stores multiple data types in a 1D group
- **Data.frame:** A two-dimensional data structure with columns and rows. Columns are variables of interest and rows represent observations.
 - Each column of a `Data.frame` can be of different data type
 - However, elements within a column must have the same data type

Tract	Population	MedAge	MedIncome	PerCapInco	PovertyRate
Tract:1807.0	4128	33.3	44422	35842	24
Tract:1718.0	5725	36.9	44865	19608	27.9
Tract:1603	5712	37	61743	23246	27.7
Tract:1816.0	5830	33.3	56875	28032	11.9
Tract:1605.0	4772	17.6	20984	11387	72.4
Tract:1817.1	5012	42	95144	38594	5.5
Tract:1708	1742	25.9	35227	15679	28
Tract:1615.0	5746	32.4	59464	23442	18.2
Tract:1602	2769	40.3	43547	26049	14.3
Tract:1218.1	6347	40.1	100668	37973	7.3
Tract:1218.0	11518	31.3	94338	32719	7.5
Tract:1913.0	1704	53.1	96971	47753	5.3
Tract:1814.0	6680	30.7	51581	31061	23
Tract:1403	3211	35	47778	20571	27.8

Note: Column 1 is a string, Column 2 Population is an integer; MedAge is a double precision

Advanced Data Structures:

The basic data types and data structures are often used to build advanced data structures. For example,

- A gray scale image is simply a matrix with additional meta data;
- A TIF image with RGB bands is an array and contain header information with regard to resolution (rows, columns, pixel length).
- GIS data use `data.frame` with additional cartographic information such as the coordinate reference system (CRS) and attribute meta data and geometry coordinates.
- Time-series data is a specialized vector or a `data.frame` with a time-stamp as the index.
- Genome data can have their own data structures.